Oracle SQL Subquery

Query Review 1

Original Query:

SELECT first_name, salary FROM employees WHERE salary > AVG(salary);

Error:

-ORA-00934: group function is not allowed here

```
SELECT first_name, salary
FROM employees
WHERE salary > AVG(salary);
```

Query Result

SQL | Executing:SELECT first_name, salaryFROM employeesWHERE salary > AVG(salary) in 0 seconds

```
ORA-00934: group function is not allowed here

https://docs.oracle.com/error-help/db/ora-00934/00934. 00000 - "group function is not allowed here"
*Cause:    One of the group functions, such as AVG, COUNT, MAX,
           MIN, SUM, STDDEV, or VARIANCE, was used in a WHERE or GROUP BY
           clause.
*Action:   Remove the group function from the WHERE or GROUP BY
           clause. The desired result may be achieved by including the
           function in a subquery or HAVING clause.
Error at Line: 6 Column: 16
```

Explanation:

-AVG is an aggregate function and cannot be used directly in the WHERE clause.

-Oracle must calculate the average salary first using a subquery.

-Then the outer query compares each employee salary with that value.

Correct Query:

SELECT first_name, salary FROM employees WHERE salary > (SELECT AVG(salary) FROM employees);

```
SELECT first_name, salary
FROM employees
WHERE salary > (SELECT AVG(salary) FROM employees);
```

Query Result

SQL | Fetched 50 rows in 0.031 seconds

	FIRST_NAME	SALARY
1	Steven	24000
2	Neena	17000
3	Lex	17000
4	Alexander	9000
5	Nancy	12008
6	Daniel	9000
7	John	8200
8	Ismael	7700
9	Jose Manuel	7800

- Type: Scalar subquery

- Business Logic: Returns employees whose salary is higher than the company average salary.

Query Review 2

Original Query:

SELECT department_name FROM departments WHERE department_id = (SELECT department_id FROM employees);

Error:

ORA-01427: single-row subquery returns more than one row

```
SELECT department_name FROM departments
WHERE department_id = ( SELECT department_id FROM employees );
```

Query Result

SQL | Executing:SELECT department_name FROM departments WHERE department_id = (SELECT department_id FROM employees) in 0 seconds

```
ORA-01427: single-row subquery returns more than one row

https://docs.oracle.com/error-help/db/ora-01427/01427. 00000 - "single-row subquery returns more than one row"
*Cause:    The outer query must use one of the keywords ANY, ALL,
           IN, or NOT IN to specify values to compare because the
           subquery returned more than one row.
*Action:   Use ANY, ALL, IN, or NOT IN to specify which values to
           compare or reword the query so only one row is retrieved.
```

Explanation:

- The subquery returns many department_id values from the employees table.

- The = operator can compare with one value only, so Oracle gives an error.

- Correct Query:

SELECT department_name FROM departments WHERE department_id IN (SELECT DISTINCT department_id FROM employees WHERE department_id IS NOT NULL);

```
SELECT department_name FROM departments WHERE department_id IN
( SELECT DISTINCT department_id FROM employees WHERE department_id IS NOT NULL);
```

Query Result

All Rows Fetched: 11 in 0.006 seconds

	DEPARTMENT_NAME
1	Administration
2	Marketing
3	Purchasing
4	Human Resources
5	Shipping
6	IT
7	Public Relations
8	Sales
9	Executive
10	Finance
11	Accounting

- Subquery Type: Multi-row Subquery

- Business Logic: Returns departments that currently have employees assigned to them.

Query Review 3

Original Query:

SELECT * FROM employees WHERE department_id IN (SELECT salary FROM employees);

Output:

The query executed but returned 0 rows.

```
SELECT * FROM employees WHERE
department_id IN ( SELECT salary FROM employees);
```

Query Result

SQL | All Rows Fetched: 0 in 0.02 seconds

EMPLOYE...	FIRST_NA...	LAST_NAME	EMAIL	PHONE_N...	HIRE_DATE	JOB_ID	SALARY	COMMISS...	MANAGER...	DEPARTM...

Explanation:

-The query compares department_id with salary.

-This logic is incorrect because department_id and salary represent different business data.

Correct Query:

SELECT * FROM employees WHERE department_id IN (SELECT department_id FROM departments);

```
SELECT * FROM employees WHERE department_id IN
( SELECT department_id FROM departments);
```

Query Result

SQL | Fetched 50 rows in 0.014 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90
4	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)	102	60
5	104	Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT_PROG	6000	(null)	103	60
6	105	David	Austin	DAUSTIN	590.423.4569	25-JUN-05	IT_PROG	4800	(null)	103	60
7	106	Valli	Pataballa	VPATABAL	590.423.4560	05-FEB-06	IT_PROG	4800	(null)	103	60
8	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-FEB-07	IT_PROG	4200	(null)	103	60
9	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100
10	109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(null)	108	100
11	110	John	Chen	JCHEN	515.124.4269	28-SEP-05	FI_ACCOUNT	8200	(null)	108	100
12	111	Ismael	Sciarra	ISCIARRA	515.124.4369	30-SEP-05	FI_ACCOUNT	7700	(null)	108	100

- Subquery Type: Multi-row Subquery

- Business Logic: Returns employees who belong to existing departments.

Query Review 4

Original Query:

SELECT first_name FROM employees WHERE salary > (SELECT salary FROM employees WHERE department_id = 90);

Error:

ORA-01427: single-row subquery returns more than one row

```
SELECT first_name FROM employees WHERE salary >
( SELECT salary FROM employees WHERE department_id = 90);
```

Query Result

SQL | Executing:SELECT first_name FROM employees WHERE salary > (SELECT salary FROM employees WHERE department_id = 90) in 0 seconds

```
ORA-01427: single-row subquery returns more than one row

https://docs.oracle.com/error-help/db/ora-01427/01427. 00000 - "single-row subquery returns more than one row"
*Cause:    The outer query must use one of the keywords ANY, ALL,
           IN, or NOT IN to specify values to compare because the
           subquery returned more than one row.
*Action:   Use ANY, ALL, IN, or NOT IN to specify which values to
           compare or reword the query so only one row is retrieved.
```

Explanation:

- The subquery returns multiple salary values from department 90.

- The > operator requires a single value for comparison.

- Therefore Oracle raises an error.

Correct Query:

SELECT first_name, salary FROM employees

WHERE salary > (SELECT AVG(salary) FROM employees

WHERE department_id = 90);

```
SELECT first_name, salary FROM employees
WHERE salary > ( SELECT AVG(salary) FROM employees
WHERE department_id = 90 );
```

Query Result

SQL | All Rows Fetched: 1 in 0.01 seconds

	FIRST_NAME	SALARY
1	Steven	24000

Subquery Type: Scalar Subquery

Business Logic: Returns employees whose salary is greater than the average salary of employees in department 90.